Results Interface Specifications

Product Name: HistoTrac

Component Name: SystemLink Results Service

Version 2.0

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1 Introduction

This specification details the structure for an outbound Results interface. This interface allows a third party system to receive a HL7 formatted message including HLA results data from HistoTrac. The Result data is recorded and formatted in HistoTrac and sent to the third party receiving system for processing.

1.1 Business Case

Third party systems that receive Results data can process the information without having to rekey the information into the third party system. HistoTrac users are currently manually entering data into the 3rd party system. Since this data has already been placed in a HistoTrac it is more efficient to pass the Result data to 3rd party through an automated interface. By utilizing the interface the result data will no longer need to be entered into 3rd party system reducing transcription errors and delays in getting paperwork processed. Result data sent through the interface in HistoTrac will be placed in a table in the HistoTrac database for future reference

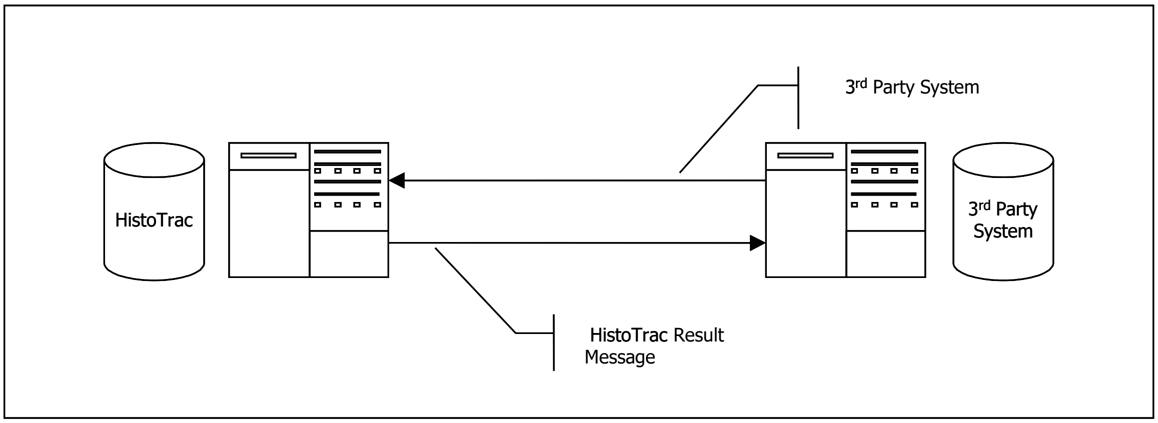
1.2 Software Change Intended Use

The Result Interface is developed so that third party software systems can receive patient/result information directly from the HistoTrac repository.

2 General Workflow

With the HistoTrac interface the third party system will receive a correctly formatted HL7 Result transaction to a specific IP Address and port. The HistoTrac HL7 Manager will be scanning the HistoTrac database for new results to send. When a new result is posted, the outbound result transaction is formatted and sent to the 3rd Party IP Address and port. When a message acknowledgement is received the next result transactions is formatted and sent. This process continues until all unprocessed results are sent to the 3rd party system.

Data Flow



3 HistoTrac and 3rd Party System Communication

3.1 Communication Protocol and Transaction Framing

All transactions will be sent using TCP/IP as the communication protocol. With the reliability of TCP, no block check calculations or data length counts are necessary. The following applies to the communication envelope for all messages. The characters listed in <> below represent ASCII characters.

- All messages and acknowledgments will begin with <11>
- All segments will end with <13>
- All messages and acknowledgments will terminate with <28><13>

3.2 Acknowledgments (ACK/NAK)

The receiving system (3rd party server) must return a communication level acknowledgment to the sending system (HistoTrac) for every message that is received. Once the server (HistoTrac) has received the acknowledgment, it may then send the next message. For example, when an result transaction is sent to the 3rd Party System, HistoTrac will wait for a return acknowledgment before sending the next message.

Acknowledgments can be either positive (ACK) or negative (NAK). For all ACK/NAK messages, the HL7 segments MSH and MSA must be returned. The MSA; 1 is valued with "CA" for ACK and with "CE" for NAK.

4 HistoTrac HL7 Result Transaction and Segments

4.1 Result Message

The Result transaction is separated into the following five segments:

MSH – Message Header
PID – Patient Identification
PV1 – Patient Visit
ORC – Common Order
OBR – Common Order Result
[{OBX}] - Result
[{NTE}] – Notes and Comments

The following table outlines the contents of each segment. Each segment is required in the Result transaction.

Legend
R – Required
O – Optional
NS – Not Supported

4.1.1 MSH – Message Header

Sequence #	Name	Max Length	Field Requirement	Notes
1	Field Separator	1	R	|
2	Encoding Characters	4	R	component delimiter (^) repeat delimiter (~) escape character (\) subcomponent delimiter (&)
3	Sending Application	50	R	
4	Sending Facility	50	O	
5	Receiving Applications	50	O	
6	Receiving Facility	50	O	
7	Date/Time of Message	12	R	CCYYMMDDHHMM
8	Security	N/A	NS	
9	Message Type	7	R	Contains message type and trigger event.
10	Message Control ID	15	R	This sequence is valued with a number that is unique to that message. The receiving system will echo this value in the MSA message.
11	Processing ID	3	R	T for Training or P for Production
12	Version ID	8	R	Specifies the HL7 version in use
13	Sequence Number	N/A	NS	
14	Continuation Pointer	N/A	NS	
15	Accept Acknowledgement Type	N/A	NS	
16	Application Acknowledgement Type	N/A	NS	
17	Country Code	N/A	NS	
18	Character Set	N/A	NS	
19	Principal Language of Message	N/A	NS	

4.1.2 PID – Patient Identification

Sequence #	Name	Max Length	Field Requirement	Notes
1	Set ID – Patient ID	N/A	NS	
2	Patient ID (External ID)	N/A	NS	

3	Patient ID (Internal ID)	20	R	Component 1 = Medical Record Number Component 2 = "" Component 3 = MulHos Code Example: Z442345^^FZQ
4	Alternate Patient ID	20	O	
5	Patient's Name	55	R	Component 1=LAST NAME Component 2=FIRST NAME Component 3=MIDDLE INITIAL Example: SMITH^JOHN^D
6	Mother's Maiden Name	40	O	
7	Date of Birth	8	O	CCYYMMDD
8	Sex	15	O	
9	Patient Alias	N/A	NS	
10	Race	15	O	
11	Patient Address	40, 40, 30, 2, 10	O	Component 1=address line 1 Component 2=address line 2 Component 3=city Component 4=state Component 5=zip code Example: 123 Oak Street ^ Suite 100 ^ Anywhere ^ VA ^ 12345
12	Country Code	15	O	
13	Phone Number – Home	15	O	
14	Phone Number – Business	15	O	
15	Language – Patient	N/A	NS	
16	Marital Status	N/A	NS	
17	Religion	N/A	NS	
18	Patient Account Number	25	O	
19	SSN Number – Patient	15	O	
20	Driver's License – Patient	N/A	NS	
21	Mother's Identifier	N/A	NS	
22	Ethnic Group	15	O	
23	Birth Place	N/A	NS	
24	Multiple Birth Indicator	N/A	NS	
25	Birth Order	N/A	NS	
26	Citizenship	N/A	NS	
27	Veteran's Military Status	N/A	NS	
28	Nationality	N/A	NS	
29	Patient Death Date/Time	N/A	NS	
30	Patient Death Indicator	N/A	NS	

4.1.3 PV1 – Patient Visit

Sequence #	Name	Max Length	Field Requirement	Notes
1	Set ID PV1	N/A	NS	
2	Patient Class	1	R	
3	Assigned Patient Location	20, 20, 20	O	Component 1 = Unit Component 2 = Room Component 3 = Bed Example: unit^room^bed
4	Admission Type	N/A	NS	
5	Pre-Admit Number	N/A	NS	
6	Prior Patient Location	N/A	NS	
7	Attending Doctor	10,	O	Component 1= Physician Code

		25, 25		Component 2 = Last Name Component 3 = First Name Example: GRT1^GRANT^JOHN
8	Referring Doctor	N/A	NS	
9	Consulting Doctor	N/A	NS	
10	Hospital Service	N/A	NS	
11	Temporary Location	N/A	NS	
12	Pre-Admit Test Indicator	N/A	NS	
13	Re-Admission Indicator	N/A	NS	
14	Admit Source	N/A	NS	
15	Ambulatory Status	N/A	NS	
16	VIP Indicators	N/A	NS	
17	Admitting Doctor	10, 25, 25	O	Component 1= Physician Code Component 2 = Last Name Component 3 = First Name Example: GRT1^GRANT^JOHN
18	Patient Type	N/A	NS	
19	Visit Number	15, 15	R	Component 1 = Episode ID Component 2 = Event ID Example: 1^2
20	Financial Class	N/A	NS	
21	Charge Price Indicator	N/A	NS	
22	Courtesy Code	N/A	NS	
23	Credit Rating	N/A	NS	
24	Contract Code	N/A	NS	
25	Contract Effective Date	N/A	NS	
26	Contract Amount	N/A	NS	
27	Contract Period	N/A	NS	
28	Interest Code	N/A	NS	
29	Transfer to Bad Debt Code	N/A	NS	
30	Transfer to Bad Debt Date	N/A	NS	
31	Bad Debt Agency Code	N/A	NS	
32	Bad Debt Transfer Amount	N/A	NS	
33	Bad Debt Recovery Amount	N/A	NS	
34	Delete Account Indicator	N/A	NS	
35	Delete Account Date	N/A	NS	
36	Discharge Disposition	N/A	NS	
37	Discharged To Location	N/A	NS	
38	Diet Type	N/A	NS	
39	Servicing Facility	N/A	NS	
40	Bed Status	N/A	NS	
41	Account Status	N/A	NS	
42	Pending Location	N/A	NS	
43	Prior Temporary Location	N/A	NS	
44	Admit Date/Time	8	R	CCYYMMDD
45	Discharge Date/Time	8	O	CCYYMMDD
46	Current Patient Balance	N/A	NS	
47	Total Charges	N/A	NS	
48	Total Adjustments	N/A	NS	
49	Total Payments	N/A	NS	
50	Alternate Visit ID	N/A	NS	
51	Visit Indicator	N/A	NS	

52	Other Health Care Provider	N/A	NS	

4.1.4 ORC – Common Order

Sequence #	Name	Max Length	Field Requirement	Notes
1	Order Control	2	R	
2	Placer Order Number	15, 50	R	Component 1 = Test Code Component 2 = Description Example: ABC^SEROLOGY CLASS I
3	Filler Order Number	15 8	O	Component 1 = Accession Number Component 2= Draw Date (CCYYMMDD) Example: 99H7242^20020802
4	Placer Order Number	N/A	NS	
5	Order Status	15	O	Order Priority
6	Response Flag	N/A	NS	
7	Quantity/Timing	N/A	NS	
8	Parent	N/A	NS	
9	Date/Time of Transaction	8	R	CCYYMMDD
10	Entered By	N/A	NS	
11	Verified By	N/A	NS	
12	Ordering Provider	10, 25, 25	O	Component 1= Physician Code Component 2 = Last Name Component 3 = First Name Example: GRT1^GRANT^JOHN
13	Enterer's Location	N/A	NS	
14	Call Back Phone Number	N/A	NS	
15	Order Effective Date/Time	N/A	NS	
16	Order Control Code Reason	N/A	NS	
17	Entering Organization	N/A	NS	
18	Entering Device	N/A	NS	
19	Action By	N/A	NS	

4.1.5 OBR – Observation Request

Sequence #	Name	Max Length	Field Requirement	Notes
1	Set ID	2	O	
2	Placer Order Number	15, 50	R	Component 1 = Test Code Component 2 = Description Example: ABC^SEROLOGY CLASS I
3	Filler Order Number	15 8	O	Component 1 = Accession Number Component 2= Draw Date (CCYYMMDD) Example: 99H7242^20020802
4	Universal Service Identifier	200	R	
5	Priority	15	O	Order Priority
6	Requested Date/Time	N/A	O	CCYYMMDD (optional - time supported)
7	Observation Date/Time	N/A	O	CCYYMMDD (optional - time supported)
8	Observation End Date/Time	N/A	NS	
9	N/A	N/A	N/A	
10	N/A	N/A	N/A	

11	Action Code	N/A	NS	
12	N/A	N/A	NS	
13	Clinical Info	N/A	O	
14	Specimen Received	N/A	O	CCYYMMDDHHMMSS
15	Specimen Source	N/A	O	
16	Ordering Provider	N/A	O	Component 1= Physician Code Component 2 = Last Name Component 3 = First Name Example: GRT1^GRANT^JOHN
17	Entering Organization	N/A	NS	
18	Placer Field	N/A	O	
19	Placer Field	N/A	O	
20	N/A	N/A	N/A	
21	N/A	N/A	N/A	
22	N/A	N/A	N/A	
23	N/A	N/A	N/A	
24	N/A	N/A	N/A	
25	Result Status	1	R	

4.1.6 OBX – Outbound Result

Sequence #	Name	Max Length	Field Requirement	Notes
1.	Set ID	N/A	NS	
2.	Value Type	N/A	NS	
3.	Test Type	15	R	Test Code
4.	Placer Order Number	N/A	NS	
5.	Result	255	R	Result
6.	Not Used	N/A	NS	
7.	Reference Range	N/A	NS	
8.	Results Abnormal	N/A	NS	
9.	Probability	N/A	NS	
10.	Abnormality Testing	N/A	NS	
11.	Corrected Flag	1	R	Y or N
12.	Date of Last Ref Change	N/A	NS	
13.	Date/Time of Observation		R	CCYYMMDD
14.	Producers ID	N/A	NS	
15.	Responsible Observer	N/A	NS	
16.	Observation Method	N/A	NS	

4.1.7 NTE – Notes and Comments

Sequence #	Name	Max Length	Field Requirement	Notes
1	Set ID – NTE	4	O	
2	Source Of Comment	8	O	
3	Comment	2000	O	

4.1.8 Sending PDFs

PDF reports created in HistoTrac can also be sent through the results interface. The PDF is encapsulated using base64 encoding and is sent in sequence 5 of the OBX segment.

An example result transaction message containing an encoded PDF is as follows:

```
MSH|^~\&|HISTO|CUST|RECVCUST|RECVCUST|20141119073252||ORU^R01|7115|T|2.3|
PID|||HS1234^^^^MRN||Last^First^E||19500101||||||||||||123456789|||||||||||
PV1|N|||UNIT^Room^Bed|||||||||
ORC|RE||||||||||||||||
```

OBR|1|22532|45-5658|||||||||||||
OBX|1|TX|Report||JVBERi0xLjMgCiXi48/TIAoxIDAgb2JqIAo8PCAKL1R5cGUgL0NhdGFsb2cgCi9QYWdlcyAyIDAgUiAKL1BhZ2VNb2Rl
IC9Vc2VOb25lIAovVmlld2VyUHJlZmVyZW5jZXMgPDwgCi9GaXRXaW5kb3cgdHJ1ZSAKL1BhZ2VMYXlvdXQgL1NpbmdsZVBhZ2UgCi9
Ob25GdWxsU2NyZWVuUGFnZU1vZGUgL1VzZU5vbmUgCj4+IAo+PiAKZW5kb2JqIAo1… (truncated to conserve space)

4.2 Message Acknowledgement

The Message Acknowledgement transaction is separated into the following two segments:

1 MSH – Message Header
2 MSA – Message Acknowledgement

The following table outlines the contents of each segment. Each segment is required in the Result transaction.

Legend
R – Required
O – Optional
NS – Not Supported

4.2.1 MSH – Message Header

Sequence #	Name	Max Length	Field Requirement	Notes
1	Field Separator	1	R	|
2	Encoding Characters	4	R	component delimiter (^) repeat delimiter (~) escape character (\) subcomponent delimiter (&)
3	Sending Application	50	R	
4	Sending Facility	50	O	
5	Receiving Applications	50	O	
6	Receiving Facility	50	O	
7	Date/Time of Message	12	R	CCYYMMDDHHMM
8	Security	N/A	NS	
9	Message Type	7	R	Contains message type and trigger event.
10	Message Control ID	15	R	This sequence is valued with a number that is unique to that message. The receiving system will echo this value in the MSA message.
11	Processing ID	3	R	T for Training or P for Production
12	Version ID	8	R	Specifies the HL7 version in use
13	Sequence Number	N/A	NS	
14	Continuation Pointer	N/A	NS	
15	Accept Acknowledgement Type	N/A	NS	
16	Application Acknowledgement Type	N/A	NS	
17	Country Code	N/A	NS	
18	Character Set	N/A	NS	
19	Principal Language of Message	N/A	NS	

4.2.2 MSA – Message Acknowledgement

Sequence #	Name	Max Length	Field Requirement	Notes
1	Acknowledgement Code	2	R	Indicates whether positive (ACK) or negative (NAK) acknowledgment. CA = ACK

				CR = NAK
2	Message Control ID	15	R	The number sent in MSH; 10 of the inbound message is echoed back to the client in this sequence.
3	Text Message	100	O	This field may be valued but only if returning a NAK and only under certain error conditions. A description of the error may be included here for troubleshooting purposes
4	Expected Sequence Number	N/A	NS	HistoTrac does not use sequence number protocol
5	Delayed ACK Type	N/A	NS	
6	Error Condition	N/A	NS	